

Computational Finance

Course WI4151

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Lecture 10 – Finite Difference Methods for Option Pricing

- The Binomial method
- Finite Difference methods for PDEs
- Solve the Black-Scholes PDE
- Binomial method as a finite difference method

Part 1 – The Binomial Method

- The Binomial method:
 - Description of the binomial method.
 - Derivation of the parameters.

Binomial method (1/3)

The binomial method uses a simple discrete model for the asset price movement under GBM model assumption.

- We let $\delta t = T/M$ denote the spacing between successive time points, where T is the time to expiry. So asset prices will be considered at times $t_i = i \cdot \delta t$, for $0 \leq i \leq M$.
- A key assumption in the binomial method is that between successive time slices the asset price moves either **up** by a factor u or **down** by a factor d .
- An upward movement occurs with probability p and a downward movement occurs with probability $1 - p$.

Binomial method (2/3)

Each node in the binomial tree represents a possible price of the underlying asset at a given point in time.

In general, at time $t_i = i \delta t$, there are $i + 1$ possible asset prices.

The n th node at the i th time step is labeled as

$$S_n^i = d^{i-n} u^n S_0, \quad 0 \leq n \leq i$$

Question: how come such a simplified model yields the same option values as BS?

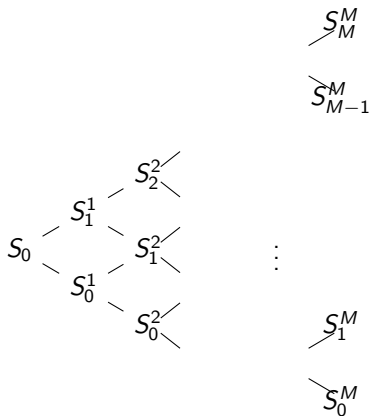


Figure: Recombining binary tree of asset prices.

Binomial method (3/3)

- In an arbitrage-free market, the option value equals the **discounted expected payoff under the risk-neutral measure**. This risk-neutral valuation principle applies in the binomial model as well.
- At the n -th node at time step i , the asset price can move to only two possible values at the next time step $i + 1$. The option value at time t_i is the risk-neutral discounted expected value of the option value at time t_{i+1} .
- To price a European-style option we perform backward induction.
 - **Initialization at maturity** $T = M \delta t$

$$V_n^M = \Lambda(S_n^M), \quad 0 \leq n \leq M.$$

- **Backward recursion** for $i = M - 1, \dots, 0$

$$V_n^i = e^{-r\delta t} (p V_{n+1}^{i+1} + (1 - p) V_n^{i+1}), \quad 0 \leq n \leq i.$$

- The option price at time 0 is obtained as V_0^0 .

Parameters in the binomial tree (1/4)

- The parameters in the binomial tree can be determined by **matching the moments** with those of the Black–Scholes model.
- Define a random variable R_i , $i \geq 1$, for each time slice:

$$R_i = \begin{cases} 1, & \text{if the asset price moves up from step } i-1 \text{ to } i, \\ 0, & \text{if the asset price moves down.} \end{cases}$$

- The asset price at step n is

$$S(n\delta t) = S_0 u^{\sum_{i=1}^n R_i} d^{n - \sum_{i=1}^n R_i}.$$

- Equivalently,

$$\ln \frac{S(n\delta t)}{S_0} = n \ln d + \ln \left(\frac{u}{d} \right) \sum_{i=1}^n R_i.$$

Parameters in the binomial tree (2/4)

- By the **Central Limit Theorem**, for large n the sum $\sum_{i=1}^n R_i$ behaves approximately like a normal random variable.
- Hence, for large n , $\ln \frac{S(n\delta t)}{S_0}$ is approximately normally distributed.
- The expectation and variance are

$$\begin{aligned}\mathbb{E}\left[\ln \frac{S(n\delta t)}{S_0}\right] &= n(p \ln u + (1-p) \ln d), \\ \text{Var}\left[\ln \frac{S(n\delta t)}{S_0}\right] &= n p(1-p) \left(\ln \frac{u}{d}\right)^2.\end{aligned}$$

- In the Black–Scholes model under the risk-neutral measure

$$\begin{aligned}\mathbb{E}\left[\ln \frac{S_t}{S_0}\right] &= \left(r - \frac{1}{2}\sigma^2\right) t, \\ \text{Var}\left[\ln \frac{S_t}{S_0}\right] &= \sigma^2 t.\end{aligned}$$

Parameters in the binomial tree (3/4)

- By **matching the expected value and variance** with those of the Black–Scholes model we obtain

$$p \ln u + (1 - p) \ln d = \left(r - \frac{1}{2}\sigma^2\right) \delta t,$$
$$\ln\left(\frac{u}{d}\right) = \frac{\sigma\sqrt{\delta t}}{\sqrt{p(1-p)}}.$$

- With $\delta t = T/M$ fixed, we have **two equations in three unknowns**:

$$p, u, d.$$

- One parameter can be chosen freely. A common choice is

$$p = \frac{1}{2}.$$

- Solving gives

$$u = e^{\sigma\sqrt{\delta t} + (r - \frac{1}{2}\sigma^2)\delta t}, \quad d = e^{-\sigma\sqrt{\delta t} + (r - \frac{1}{2}\sigma^2)\delta t}.$$

Parameters in the binomial tree (4/4)

- The arguments above suggest that the binomial asset price model approximates the Black–Scholes model when δt is small (i.e. when the number of steps M is large).
- For a fixed maturity T , the binomial model converges to the Black–Scholes model as $M \rightarrow \infty$.
- Consequently, the binomial option price converges to the Black–Scholes price.

Convergence of binomial method

The convergence is not monotonic, but it is possible to show that the convergence rate is of first order:

- The absolute value of the error in the binomial method approximation with M time steps can be bounded by K/M , with some constant K .

M	diff
100	-0.0012
200	0.0034
400	-0.0002
BS val.	1.4728

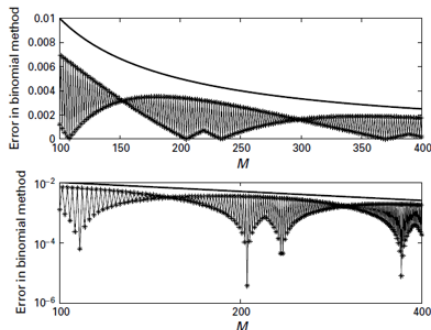


Fig. 16.3. Upper picture: Error in the binomial method for a European put as the number of time points, M , increases from 100 to 400. Solid line is $1/M$. Lower picture: same data on a log-log scale.

Source: An Introduction to Financial Option Valuation: Mathematics, Stochastics and Computation, by Desmond J. Higham, Cambridge University Press, 2004.

Pros:

- Binomial method is straightforward to describe and implement.
- It has the advantage that it is readily adapted to a range of non-European options for which no analytical formula is available.
- In particular, the binomial method provides the simplest means to value American options.

Cons:

- It is only suitable for GBM model assumption, or, more generally and from theoretical point of view, for model assumptions whereby the first three moments of the underlying process drive the option values.
- The convergence is not regular.

Part 2 – Finite Difference Methods

- The Finite Difference methods:
 - The Heat Equation
 - Finite Difference operators
 - FTCS and BTCS
 - Local accuracy
 - Von Neumann stability
 - Convergence
 - Crank-Nicholson method

The Heat Equation (1/2)

Let $u(x, t)$ be the temperature variable depending on variables $x \in [0, L]$, and $t \in [0, T]$ denoting 1-D space grid and time respectively.

Then, the PDE

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$

with BCs,

$$u(0, t) = a(t), \quad u(L, t) = b(t)$$

and the IC,

$$u(x, 0) = g(x)$$

is known as the heat equation.

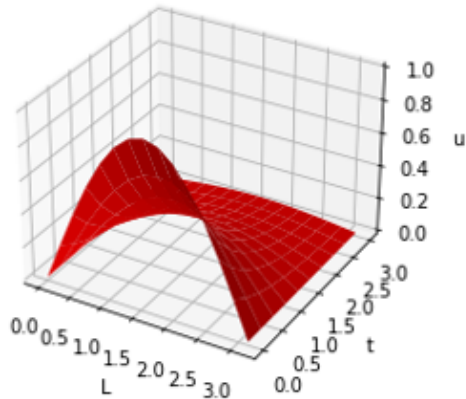
The Heat Equation (2/2)

Analytical solution exists in some cases and can be solved for instance, using the method for separation of variables.

EXAMPLE: $L = \pi$, $g(x) = \sin x$, $a(t) = b(t) = 0$,

$$u(x, t) = e^{-t} \sin(x)$$

To obtain an approximate solution for a general heat equation, with any given boundary and initial conditions, we use the help of numerical methods, like the **Finite Difference method**.



Finite Difference operators

$y : \mathbb{R} \rightarrow \mathbb{R}$, where y is smooth,

$$\frac{y(x+h) - y(x)}{h} \approx \frac{dy}{dx}(x)$$

Using Taylor's series for $y_m = y(mh)$ and taking the difference and $m \in \mathbb{Z}$,

$$y_{m+1} - y_m = h y' + \frac{1}{2} h^2 y'' + \dots$$

For example, the forward difference operator is defined as,

$$\Delta y_m = y_{m+1} - y_m$$

Different Difference operators

Table: Difference operators

Operator	Symbol	Definition	Taylor series
Forward difference	Δ	$y_{m+1} - y_m$	$hy' + \frac{1}{2}h^2y'' + \dots$
Backward difference	∇	$y_m - y_{m-1}$	$hy' - \frac{1}{2}h^2y'' + \dots$
Half central difference	δ	$y_{m+\frac{1}{2}} - y_{m-\frac{1}{2}}$	$hy' - \frac{1}{24}h^3y''' + \dots$
Full central difference	Δ_0	$\frac{1}{2}(y_{m+1} - y_{m-1})$	$hy' + \frac{1}{6}h^3y''' + \dots$
Second order central difference	δ^2	$y_{m+1} - 2y_m + y_{m-1}$	$h^2y'' - \frac{1}{12}h^4y'''' + \dots$
Shift	E	y_{m+1}	$y + hy' + \dots$
Average	μ	$\frac{1}{2}\left(y_{m+\frac{1}{2}} + y_{m-\frac{1}{2}}\right)$	$y + \frac{1}{8}h^2y'' + \dots$

Discretization (1/2)

- The first step is to discretize the given PDE
- The approximate solution is computed at these finite set of points
- The space axis can be divided into $N_x + 1$ equally spaced points, jh , $j = 0, \dots, N_x$, where $h = L/N_x$
- Similarly, time axis is subdivided into $N_t + 1$ equally spaced points ik , $i = 0, \dots, N_t$, where $k = T/N_t$
- The points (jh, ik) form **the grid, or the mesh**
- We seek solution U_j^i that approximates the solution on the grid as,

$$U_j^i \approx u(jh, ik), \quad 0 \leq j \leq N_x \text{ and } 0 \leq i \leq N_t$$

- Note: The choice of discretization grid for the space axis can also be unequally spaced, in certain cases

Discretization (2/2)

The heat equation involves two independent variables, x and t .

The choice for either taking the forward step or the previous step or a mixture of both for approximating the derivative can be classified as,

- Forward Differencing
- Backward Differencing
- Central Differencing

In the case for forward difference, discretizing along t can be denoted as,

$$\Delta_t U_j^i = U_j^{i+1} - U_j^i$$

Similarly, along x ,

$$\Delta_x U_j^i = U_{j+1}^i - U_j^i$$

FTCS – **F**orward Difference in **T**ime, **C**entral Difference in **S**pace

$$\frac{U_j^{i+1} - U_j^i}{k} = \frac{U_{j+1}^i - 2U_j^i + U_{j-1}^i}{h^2}$$

$$\Rightarrow U_j^{i+1} = \nu U_{j+1}^i + (1 - 2\nu) U_j^i + \nu U_{j-1}^i$$

where $\nu = k/h^2$ is the mesh ratio.

We already have from BCs,

$$U_0^{i+1} = a((i+1)k) \quad \text{and} \quad U_{N_x}^{i+1} = b((i+1)k)$$

From the IC,

$$U_j^0 = g(jh)$$

FTCS: from the stencil to a linear system (1/3)

For the interior nodes $j = 1, \dots, N_x - 1$, the FTCS update is

$$U_j^{i+1} = \nu U_{j+1}^i + (1 - 2\nu)U_j^i + \nu U_{j-1}^i.$$

Write out the first few equations explicitly:

$$U_1^{i+1} = (1 - 2\nu)U_1^i + \nu U_2^i + \nu U_0^i,$$

$$U_2^{i+1} = \nu U_1^i + (1 - 2\nu)U_2^i + \nu U_3^i,$$

$\vdots$

$$U_{N_x-2}^{i+1} = \nu U_{N_x-3}^i + (1 - 2\nu)U_{N_x-2}^i + \nu U_{N_x-1}^i,$$

$$U_{N_x-1}^{i+1} = \nu U_{N_x-2}^i + (1 - 2\nu)U_{N_x-1}^i + \nu U_{N_x}^i.$$

The boundary values U_0^i and $U_{N_x}^i$ are known from the boundary conditions, so only

$$U_1^i, \dots, U_{N_x-1}^i$$

are unknowns.

FTCS: from the stencil to a linear system (2/3)

Collect the interior unknowns into the vector

$$\mathbf{U}^i = (U_1^i, U_2^i, \dots, U_{N_x-1}^i)^T.$$

Then the previous equations can be written as

$$\begin{pmatrix} U_1^{i+1} \\ U_2^{i+1} \\ \vdots \\ U_{N_x-2}^{i+1} \\ U_{N_x-1}^{i+1} \end{pmatrix} = \begin{pmatrix} 1-2\nu & \nu & 0 & \cdots & 0 \\ \nu & 1-2\nu & \nu & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \nu & 1-2\nu & \nu \\ 0 & \cdots & 0 & \nu & 1-2\nu \end{pmatrix} \begin{pmatrix} U_1^i \\ U_2^i \\ \vdots \\ U_{N_x-2}^i \\ U_{N_x-1}^i \end{pmatrix} + \begin{pmatrix} \nu U_0^i \\ 0 \\ \vdots \\ 0 \\ \nu U_{N_x}^i \end{pmatrix}.$$

Using the boundary conditions

$$U_0^i = a(ik), \quad U_{N_x}^i = b(ik),$$

we obtain

$$\mathbf{U}^{i+1} = F \mathbf{U}^i + \mathbf{p}^i.$$

FTCS: from the stencil to a linear system (3/3)

FTCS – **F**orward Difference in **T**ime, **C**entral Difference in **S**pace

$$\mathbf{U}^{i+1} = F \mathbf{U}^i + \mathbf{p}^i, \quad 0 \leq i \leq N_t - 1$$

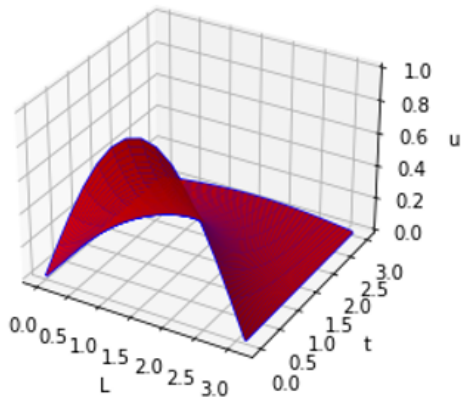
where

$$\mathbf{U}^i := \begin{pmatrix} U_1^i \\ U_2^i \\ \vdots \\ U_{N_x-1}^i \end{pmatrix}_{(N_x-1) \times 1}, \quad \mathbf{U}^0 := \begin{pmatrix} g(h) \\ g(2h) \\ \vdots \\ g((N_x-1)h) \end{pmatrix}_{(N_x-1) \times 1}, \quad \mathbf{p}^i = \begin{pmatrix} \nu a(ik) \\ 0 \\ \vdots \\ 0 \\ \nu b(ik) \end{pmatrix}_{(N_x-1) \times 1}$$

$$F = \begin{pmatrix} 1-2\nu & \nu & 0 & \cdots & 0 \\ \nu & 1-2\nu & \nu & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & 1-2\nu & \nu \\ 0 & \cdots & 0 & \nu & 1-2\nu \end{pmatrix}_{(N_x-1) \times (N_x-1)}$$

Example:

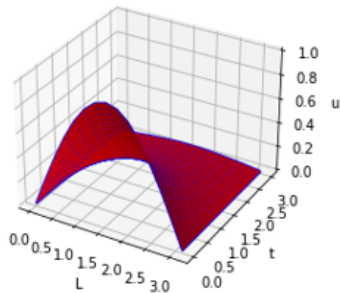
$\nu = 0.299381$ $\text{inf_norm_error} = 0.001234$



Stability and convergence in FTCS

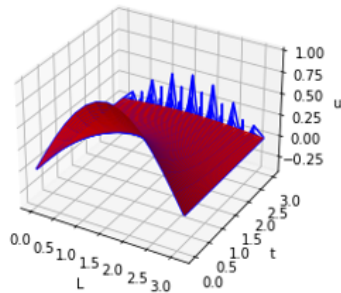
$$\nu \approx 0.3$$

$\nu = 0.299381$ inf_norm_error= 0.001234



$$\nu \approx 0.6$$

$\nu = 0.633796$ inf_norm_error= 0.460958



FTCS is stable when $\nu \leq 0.5$

BTCS – **B**ackward Difference in **T**ime, **C**entral Difference in **S**pace

$$\frac{U_j^i - U_j^{i-1}}{k} = \frac{U_{j+1}^i - 2U_j^i + U_{j-1}^i}{h^2}$$

$$\Rightarrow U_j^{i+1} = U_j^i + \nu(U_{j+1}^{i+1} - 2U_j^{i+1} + U_{j-1}^{i+1})$$

In BTCS, there is no explicit way to compute $\{U_j^{i+1}\}_{j=1}^{N_x-1}$ from U_j^i .

BTCS – **B**ackward Difference in **T**ime, **C**entral Difference in **S**pace

$$B \mathbf{U}^{i+1} = \mathbf{U}^i + \mathbf{q}^i, \quad 0 \leq i \leq N_t - 1$$

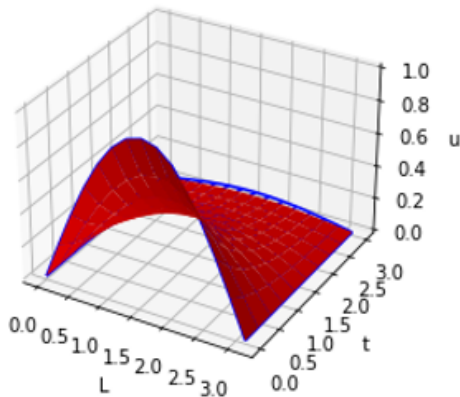
where

$$\mathbf{q}^i = \begin{pmatrix} \nu a((i+1)k) \\ 0 \\ \vdots \\ 0 \\ \nu b((i+1)k) \end{pmatrix}_{(N_x-1) \times 1}$$

$$B = \begin{pmatrix} 1+2\nu & -\nu & 0 & \cdots & 0 \\ -\nu & 1+2\nu & -\nu & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & 1+2\nu & -\nu \\ 0 & \cdots & 0 & -\nu & 1+2\nu \end{pmatrix}_{(N_x-1) \times (N_x-1)}$$

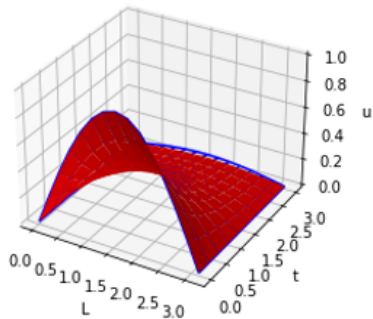
Example:

$\nu = 6.619651$ $\text{inf_norm_error} = 0.055324$

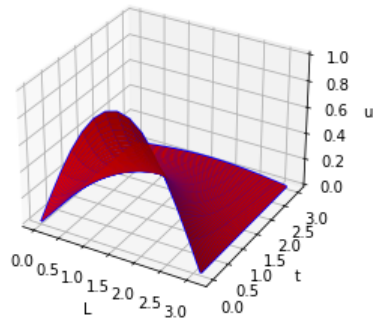


Stability and convergence in BTCS

$\nu = 6.619651$ inf_norm_error= 0.055324



$\nu = 0.633796$ inf_norm_error= 0.007314



BTCS is unconditionally stable: $\nu > 0$.

Local Accuracy

How do we judge the accuracy of a Finite Difference method?

Residual: exact solution is substituted into the difference formula,

$$R_j^i := k^{-1} \Delta_t u_j^i - h^{-2} \delta_x^2 u_j^i$$

For FTCS,

$$\begin{aligned} R_j^i &= \frac{\partial u}{\partial t} + \frac{1}{2}k \frac{\partial^2 u}{\partial t^2} + O(k^2) - \frac{\partial^2 u}{\partial x^2} - \frac{1}{12}h^2 \frac{\partial^4 u}{\partial x^4} - O(h^4) \\ &= \frac{1}{2}k \frac{\partial^2 u}{\partial t^2} - \frac{1}{12}h^2 \frac{\partial^4 u}{\partial x^4} + O(k^2) + O(h^4) \end{aligned}$$

$\Rightarrow$ FTCS is first order in time $O(k)$ and second order in space $O(h^2)$.

For BTCS,

$$R_j^i = -\frac{1}{2}k \frac{\partial^2 u}{\partial t^2} - \frac{1}{12}h^2 \frac{\partial^4 u}{\partial x^4} + O(k^2) + O(h^4)$$

$\Rightarrow$ BTCS is also first order in time $O(k)$ and second order in space $O(h^2)$.

Where does truncation error come from?

- Consider the heat equation $u_t = u_{xx}$.
- The FTCS scheme approximates it by

$$\frac{U_j^{i+1} - U_j^i}{k} = \frac{U_{j+1}^i - 2U_j^i + U_{j-1}^i}{h^2}.$$

- Substitute the **exact solution** $u(x_j, t_i)$ into the stencil

$$R_j^i = \frac{u(x_j, t_i + k) - u(x_j, t_i)}{k} - \frac{u(x_j + h, t_i) - 2u(x_j, t_i) + u(x_j - h, t_i)}{h^2}.$$

- Using Taylor expansions

$$\frac{u(x, t + k) - u(x, t)}{k} = u_t + \frac{1}{2}ku_{tt} + O(k^2),$$

$$\frac{u(x + h, t) - 2u(x, t) + u(x - h, t)}{h^2} = u_{xx} + \frac{1}{12}h^2u_{xxxx} + O(h^4).$$

- Since $u_t = u_{xx}$, the leading terms cancel.

Accuracy of FTCS and BTCS

- From the truncation error

$$R_j^i = \frac{1}{2}k u_{tt} - \frac{1}{12}h^2 u_{xxxx} + O(k^2) + O(h^4),$$

the FTCS scheme has

FTCS: $O(k)$ accuracy in time, $O(h^2)$ in space.

- For the BTCS scheme

$$\frac{U_j^{i+1} - U_j^i}{k} = \frac{U_{j+1}^{i+1} - 2U_j^{i+1} + U_{j-1}^{i+1}}{h^2},$$

the Taylor expansion gives

$$R_j^i = -\frac{1}{2}k u_{tt} - \frac{1}{12}h^2 u_{xxxx} + O(k^2) + O(h^4).$$

- Therefore,

BTCS: $O(k)$ in time, $O(h^2)$ in space.

Von Neumann stability test

- **Convergence:** Error must tend to zero as the grid size tends to zero.
- **Lax equivalence theorem** states that a method converges if and only if its local accuracy tends to zero as grid size decreases and satisfies a stability condition.
- **Stability:** A finite difference method generating approximations U_j^i is stable in the sense of von Neumann if, ignoring initial and boundary conditions, under the substitution

$$U_j^i = \xi^i e^{\mathbf{i}\beta jh}$$

it follows that $|\xi| \leq 1$ for all $\beta h \in [-\pi, \pi]$. Here $\mathbf{i}$ denotes the unit imaginary number.

How does the Von Neumann test work? (1/6)

- **Goal:** Determine whether numerical errors grow over time.
- Suppose the numerical solution contains an error e_j^i .
- Key question:

If a small error appears, will it grow as time advances?

- Stability requires

$$|e_j^i| \not\rightarrow \infty \quad \text{as } i \rightarrow \infty.$$

- Instead of studying an arbitrary error function directly, we analyze how the scheme treats **simple wave components**.

How does the Von Neumann test work? (2/6)

- Consider the FTCS scheme

$$U_j^{i+1} = \nu U_{j+1}^i + (1 - 2\nu)U_j^i + \nu U_{j-1}^i.$$

- Let the exact solution on the grid be u_j^i .
- Define the error

$$e_j^i = U_j^i - u_j^i.$$

- Subtracting the schemes applied to U and u gives

$$e_j^{i+1} = \nu e_{j+1}^i + (1 - 2\nu)e_j^i + \nu e_{j-1}^i + R_j^i.$$

- Key point:** the same finite difference operator propagates the error.

How does the Von Neumann test work? (3/6)

- Any grid function can be written as a sum of **Fourier modes**

$$e_j^i = \sum_{\beta} \hat{e}_{\beta}^i e^{i\beta jh}.$$

- This is the discrete Fourier representation of the error.
- Interpretation:

$$\text{error} = \text{wave}_1 + \text{wave}_2 + \text{wave}_3 + \cdots$$

- Therefore the behaviour of the error can be understood by studying how the scheme acts on each wave component.

How does the Von Neumann test work? (4/6)

- Finite difference schemes such as FTCS are
 - linear
 - shift-invariant in space
- Because the scheme is linear, i.e., $F\left(\sum_{\beta} \hat{e}_{\beta}^i e^{i\beta jh}\right) = \sum_{\beta} F\left(\hat{e}_{\beta}^i e^{i\beta jh}\right)$, each Fourier mode evolves independently.
- Because the scheme is shift-invariant, a Fourier mode remains a Fourier mode. For example, applying the FTCS stencil gives

$$\begin{aligned} F(e^{i\beta jh}) &= \nu e^{i\beta(j+1)h} + (1 - 2\nu)e^{i\beta jh} + \nu e^{i\beta(j-1)h}. \\ &= \left(\nu e^{i\beta h} + (1 - 2\nu) + \nu e^{-i\beta h}\right) e^{i\beta jh} = \xi(\beta) e^{i\beta jh}. \end{aligned}$$

- Therefore, there are **no cross terms between modes**.

How does the Von Neumann test work? (5/6)

- Since each mode evolves independently, we study a single Fourier mode

$$U_j^i = \xi(\beta)^i e^{i\beta jh}.$$

- Substitute this expression into the FTCS scheme

$$U_j^{i+1} = \nu U_{j+1}^i + (1 - 2\nu)U_j^i + \nu U_{j-1}^i.$$

- Using the Fourier mode,

$$\xi(\beta)^{i+1} e^{i\beta jh} = \nu \xi(\beta)^i e^{i\beta(j+1)h} + (1 - 2\nu)\xi(\beta)^i e^{i\beta jh} + \nu \xi(\beta)^i e^{i\beta(j-1)h}.$$

- Dividing by $\xi(\beta)^i e^{i\beta jh}$ gives

$$\xi(\beta) = \nu e^{i\beta h} + (1 - 2\nu) + \nu e^{-i\beta h}.$$

- This quantity is called the **amplification factor**.
- Interpretation: each time step multiplies the wave amplitude by $\xi(\beta)$

$$U_j^{i+1} = \xi(\beta)U_j^i.$$

How does the Von Neumann test work? (6/6)

- After one time step the error becomes

$$e_j^{i+1} = \sum_{\beta} \xi(\beta) \hat{e}_{\beta}^i e^{i\beta jh}.$$

- After n time steps

$$e_j^{i+n} = \sum_{\beta} \xi(\beta)^n \hat{e}_{\beta}^i e^{i\beta jh}.$$

- **Stability condition**

$$|\xi(\beta)| \leq 1 \quad \text{for all } \beta h \in [-\pi, \pi].$$

- If every Fourier mode remains bounded, the entire error remains bounded.

Calculating the stability condition for FTCS

Substituting $U_j^i = \xi^i e^{i\beta jh}$ into the FTCS formula,

$$\xi^{i+1} e^{i\beta jh} = \nu \xi^i e^{i\beta jh} e^{i\beta h} + (1 - 2\nu) \xi^i e^{i\beta jh} + \nu \xi^i e^{i\beta jh} e^{-i\beta h}.$$

So

$$\begin{aligned}\bar{\xi} &= \nu e^{i\beta h} + (1 - 2\nu) + \nu e^{-i\beta h} = 1 + \nu (e^{i\beta h} - 2 + e^{-i\beta h}) \\ &= 1 + \nu (e^{i\frac{1}{2}\beta h} - e^{-i\frac{1}{2}\beta h})^2 = 1 + \nu (2i \sin(\frac{1}{2}\beta h))^2 \\ &= 1 - 4\nu \sin^2(\frac{1}{2}\beta h).\end{aligned}$$

The condition $|\bar{\xi}| \leq 1$ thus becomes

$$|1 - 4\nu \sin^2(\frac{1}{2}\beta h)| \leq 1,$$

which simplifies to

$$0 \leq \nu \sin^2(\frac{1}{2}\beta h) \leq \frac{1}{2}.$$

Von Neumann stability analysis for BTCS

BTCS scheme:

$$\frac{U_j^{i+1} - U_j^i}{k} = \alpha \frac{U_{j+1}^{i+1} - 2U_j^{i+1} + U_{j-1}^{i+1}}{h^2}.$$

Let $\nu = \frac{\alpha k}{h^2}$. and substitute the Fourier mode $U_j^i = \xi^i e^{i\beta jh}$. to yield

$$\xi^{i+1} e^{i\beta jh} - \xi^i e^{i\beta jh} = \nu \xi^{i+1} e^{i\beta jh} (e^{i\beta h} - 2 + e^{-i\beta h}).$$

Dividing by $\xi^i e^{i\beta jh}$ gives

$$\xi - 1 = \nu \xi (e^{i\beta h} - 2 + e^{-i\beta h}).$$

Using

$$e^{i\beta h} - 2 + e^{-i\beta h} = -4 \sin^2 \left(\frac{\beta h}{2} \right),$$

we obtain

$$\xi = \frac{1}{1 + 4\nu \sin^2 \left(\frac{\beta h}{2} \right)}.$$

Crank-Nicholson (1/3)

$$k^{-1} \Delta_t U_j^{i+0.5} = h^{-2} \delta_x^2 U_j^{i+0.5}$$

Points $i + 0.5$ are not present on the grid. So we apply the time averaging operator μ_t :

$$k^{-1} \Delta_t U_j^{i+0.5} = h^{-2} \delta_x^2 \mu_t U_j^{i+0.5}$$

$$k^{-1} (U_j^{i+1} - U_j^i) = h^{-2} \delta_x^2 \left(\frac{1}{2} U_j^{i+1} + U_j^i \right)$$

Local accuracy is of second order both in time and space:

$$\sim O(k^2) + O(h^2)$$

Final expression:

$$2(1 + \nu)U_j^{i+1} = \nu U_{j+1}^{i+1} + \nu U_{j-1}^{i+1} + \nu U_{j-1}^i + U_{j+1}^i + U_{j-1}^i + 2(1 - \nu)U_j^i$$

In terms of matrix-vector product:

$$B \mathbf{U}^{i+1} = F \mathbf{U}^i + \mathbf{r}^i, \quad 0 \leq i \leq N_t - 1$$

with

$$B = \begin{pmatrix} 1 + \nu & -\frac{1}{2}\nu & 0 & \cdots & 0 \\ -\frac{1}{2}\nu & 1 + \nu & -\frac{1}{2}\nu & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -\frac{1}{2}\nu & 1 + \nu & -\frac{1}{2}\nu \\ 0 & \cdots & 0 & -\frac{1}{2}\nu & 1 + \nu \end{pmatrix}_{(N_x-1) \times (N_x-1)}$$

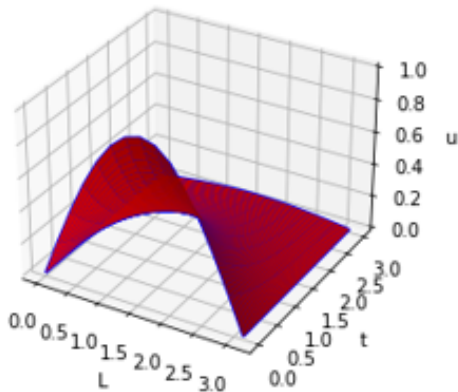
$$F = \begin{pmatrix} 1 - \nu & \frac{1}{2}\nu & 0 & \cdots & 0 \\ \frac{1}{2}\nu & 1 - \nu & \frac{1}{2}\nu & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & \frac{1}{2}\nu & 1 - \nu & \frac{1}{2}\nu \\ 0 & \cdots & 0 & \frac{1}{2}\nu & 1 - \nu \end{pmatrix}_{(N_x-1) \times (N_x-1)}$$

$$\mathbf{r}^i = \begin{pmatrix} \frac{1}{2}\nu(a(ik) + a((i+1)k)) \\ 0 \\ \vdots \\ 0 \\ \frac{1}{2}\nu(b(ik) + b((i+1)k)) \end{pmatrix}_{(N_x-1) \times 1}$$

Crank-Nicholson vs BTCS

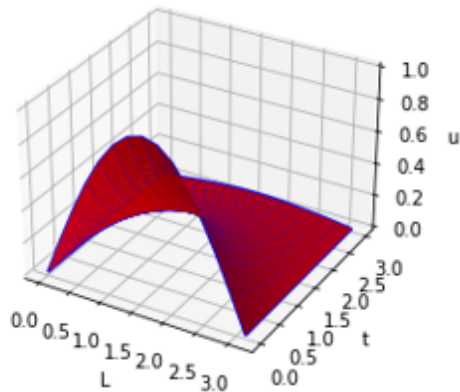
Crank-Nicholson

$\nu = 0.633796$ inf_norm_error= 0.001513



BTCS

$\nu = 0.633796$ inf_norm_error= 0.007314



Part 3 – Solve the BS PDE

- The Binomial method:
 - Black-Scholes PDE in reverse time
 - Initial and boundary conditions
 - FTCS, BTCS and Crank-Nicholson
 - Binomial method as a finite difference method

Black-Scholes PDE in reverse time

The Black-Scholes PDE includes a final time condition irrespective the option type.

Make a change of variable $\tau = T - t$ to specify the problem in initial time condition form.

$$\frac{\partial V}{\partial \tau} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rS \frac{\partial V}{\partial S} + rV = 0$$

We focus on European calls and puts. Domain $S \in [0, \infty)$ is impractical for a numerical method, hence can be truncated at a reasonably large value L .

For a European **call**, at $t = T \Rightarrow \tau = 0$:

$$V(S, 0) = \max(S_0 - E, 0), \quad V(0, \tau) = 0, \quad V(L, \tau) = L - E e^{-r\tau}$$

For a European **put**, at $\tau = 0$:

$$V(S, 0) = \max(E - S(0), 0), \quad V(0, \tau) = E e^{-r\tau}, \quad V(L, \tau) = 0$$

Discretization

into $N_x + 1$ equally spaced points and t axis into $N_t + 1$ equally spaced points by the grid (jh, ik) , $j = 0, \dots, N_x$, $i = 0, \dots, N_t$.

Let $\mathbf{V}^i := \begin{pmatrix} V_1^i \\ V_2^i \\ \vdots \\ V_{N_x-1}^i \end{pmatrix}$ denote the numerical solution of option price $\mathbf{V}$ at time i .

The initial and boundary conditions:

$$V_j^0 = \begin{cases} \max(S_0 - E, 0), & \text{European Put Option} \\ \max(E - S_0, 0), & \text{European Call Option} \end{cases} \quad \forall j$$

$$V_0^{N_t} = \begin{cases} 0, & \text{European Put Option} \\ E e^{-r\tau}, & \text{European Call Option} \end{cases}$$

$$V_{N_x}^{N_t} = \begin{cases} L - E e^{-r\tau}, & \text{European Put Option} \\ 0, & \text{European Call Option} \end{cases}$$

FTCS method

Discretization of the Black-Scholes PDE using the FTCS finite difference scheme yields

$$\frac{V_j^{i+1} - V_j^i}{k} - \frac{1}{2}\sigma^2(jh)^2 \frac{V_{j+1}^i - 2V_j^i + V_{j-1}^i}{h^2} - r(jh) \frac{V_{j+1}^i - V_{j-1}^i}{2h} + r V_j^i = 0$$

Rearranging the terms to obtain

$$V_j^{i+1} = (1 - rk)V_j^i + \frac{1}{2}k\sigma^2 j^2 (V_{j+1}^i - 2V_j^i + V_{j-1}^i) + \frac{1}{2}krj(V_{j+1}^i - V_{j-1}^i)$$

Represented by matrix-vector product:

$$\mathbf{V}^{i+1} = F \mathbf{V}^i + \mathbf{p}^i, \quad 0 \leq i \leq N_t - 1$$

where $F = (1 - rk)I + \frac{1}{2}k\sigma^2 D_2 T_2 + \frac{1}{2}kr D_1 T_1$ and

$$\mathbf{p}^i = \begin{pmatrix} \frac{1}{2}k(\sigma^2 - r)V_0^i \\ 0 \\ \vdots \\ 0 \\ \frac{1}{2}k(N_x - 1)(\sigma^2(N_x - 1) + r)V_{N_x}^i \end{pmatrix}$$

BTCS method

Discretizing similarly using the BTCS finite difference scheme yields,

$$\frac{V_j^{i+1} - V_j^i}{k} - \frac{1}{2}\sigma^2(jh)^2 \frac{V_{j+1}^{i+1} - 2V_j^{i+1} + V_{j-1}^{i+1}}{h^2} - r(jh) \frac{V_{j+1}^{i+1} - V_{j-1}^{i+1}}{2h} + r V_j^{i+1} = 0$$

Rearranging the terms to obtain

$$V_j^{i+1} + rk V_j^{i+1} - \frac{1}{2}k\sigma^2 j^2 (V_{j+1}^{i+1} - 2V_j^{i+1} + V_{j-1}^{i+1}) - \frac{1}{2}krj (V_{j+1}^{i+1} - V_{j-1}^{i+1}) = V_j^i$$

The update step can be written as,

$$B \mathbf{V}^{i+1} = \mathbf{V}^i + \mathbf{q}^i, \quad 0 \leq i \leq N_t - 1$$

with $B = (1 + rk)I - \frac{1}{2}k\sigma^2 D_2 T_2 - \frac{1}{2}kr D_1 T_1$ and

$$\mathbf{q}^i = \begin{pmatrix} \frac{1}{2}k(\sigma^2 - r)V_0^{i+1} \\ 0 \\ \vdots \\ 0 \\ \frac{1}{2}k(N_x - 1)(\sigma^2(N_x - 1) + r)V_{N_x}^{i+1} \end{pmatrix}$$

FTCS, BTCS and Crank-Nicholson for Black-Scholes

$$D_1 = \begin{bmatrix} 1 & 0 & \cdots & \cdots & 0 \\ 0 & 2 & 0 & \ddots & \vdots \\ \vdots & 0 & 3 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & N_x - 1 \end{bmatrix}, \quad D_2 = \begin{bmatrix} 1^2 & 0 & \cdots & \cdots & 0 \\ 0 & 2^2 & 0 & \ddots & \vdots \\ \vdots & 0 & 3^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & (N_x - 1)^2 \end{bmatrix}$$

$$T_1 = \begin{bmatrix} 0 & 1 & 0 & \cdots & \cdots & 0 \\ -1 & 0 & 1 & \ddots & & \vdots \\ 0 & \ddots & \ddots & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & \ddots & 0 \\ \vdots & & & -1 & 0 & 1 \\ 0 & \cdots & \cdots & 0 & -1 & 0 \end{bmatrix}, \quad T_2 = \begin{bmatrix} -2 & 1 & 0 & \cdots & \cdots & 0 \\ 1 & -2 & 1 & \ddots & & \vdots \\ 0 & 1 & \ddots & \ddots & & \vdots \\ \vdots & & \ddots & \ddots & 1 & 0 \\ \vdots & & & 1 & -2 & 1 \\ 0 & \cdots & \cdots & 0 & 1 & -2 \end{bmatrix}.$$

Crank-Nicholson

If we take the average of the FTCS and the BTCS formulas, we obtain the Crank-Nicholson scheme

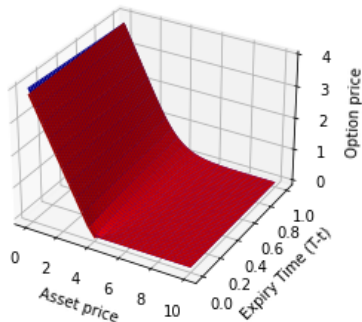
$$\left(\frac{1}{2}I + B\right) \mathbf{V}^{i+1} = \left(\frac{1}{2}I + F\right) \mathbf{V}^i + \frac{1}{2}(\mathbf{p}^i + \mathbf{q}^i)$$

FTCS, BTCS and Crank-Nicholson for Black-Scholes (1/2)

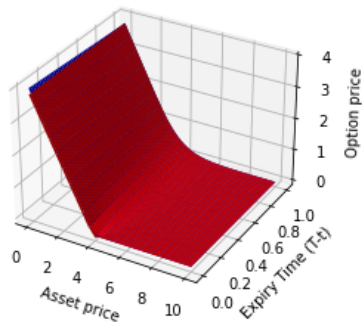
Example: European put option with $E = 4$, $\sigma = 0.3$, $r = 0.03$ and $T = 1$. Asset range is truncated at $L = 10$. $N_x = 50$ and $N_t = 500$. We compared our solution with maximum error at time $t = 0$, which is defined as

$$\epsilon_0 = \max_{1 \leq j \leq N_x - 1} |V_j^{N_t} - V(jh, \tau = T)|$$

FTCS inf_norm_error at t=0 : 0.001507



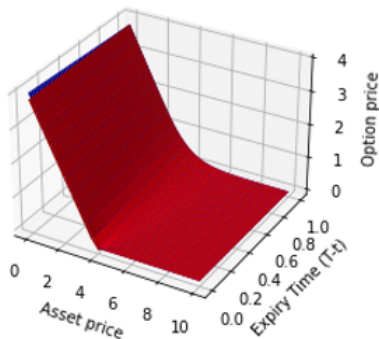
BTCS inf_norm_error at t=0 : 0.001726



FTCS, BTCS and Crank-Nicholson for Black-Scholes (2/2)

Example: European put option with $E = 4$, $\sigma = 0.3$, $r = 0.03$ and $T = 1$. Asset range is truncated at $L = 10$. $N_x = 50$ and $N_t = 500$.

Crank Nicholson inf_norm_error at t=0 : 0.001617



Binomial Method as finite differences (1/3)

Similarities between the binomial and finite difference methods:

- works with the time and asset domains discretization,
- Stepping forward along time axis
- Improves accuracy as grid is made finer.
- Finite difference methods are more general, in that they produce option values at all grid-points $\{jh, ik\}$
- At time zero, option values are available for all initial asset prices $0, h, 2h, \dots, L$.
- The binomial method may be regarded as some explicit finite difference scheme that has been customized to produce a single time-zero option value.

Binomial Method as finite differences (2/3)

$$\frac{\partial V}{\partial \tau} - \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rS \frac{\partial V}{\partial S} + rV = 0$$

Take $X = \ln S$,

$$\frac{\partial V}{\partial \tau} - \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial X^2} - \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial V}{\partial X} - rV = 0$$

Let $V = e^{rt}W$, to eliminate the linear term in V :

$$\frac{\partial W}{\partial \tau} - \frac{1}{2}\sigma^2 \frac{\partial^2 W}{\partial X^2} - \left(r - \frac{1}{2}\sigma^2\right) \frac{\partial W}{\partial X} = 0$$

Binomial Method as finite differences (3/3)

Applying BTCS to have

$$\frac{W_j^{i+1} - W_j^i}{k} + \frac{\sigma^2}{2} \frac{W_{j+1}^{i+1} - 2W_j^{i+1} + W_{j-1}^{i+1}}{h^2} + \left(r - \frac{1}{2}\sigma^2\right) \frac{W_{j+1}^{i+1} - W_{j-1}^{i+1}}{2h} = 0$$

Setting $h^2 = \sigma^2 k$, to eliminate W_j^{i+1} terms,

$$W_j^i = p^* W_{j+1}^{i+1} + (1 - p^*) W_{j-1}^{i+1}$$

where $p^* = \frac{1}{2} \left(1 + \sqrt{\frac{k}{\sigma}} \cdot r - \sigma^2 \right).$

Transforming back to V to yield

$$V_j^i = e^{-rk} (p^* V_{j+1}^{i+1} + (1 - p^*) V_{j-1}^{i+1})$$

References

- LeVeque (2007), Finite Difference Methods for ODEs and PDEs.
- An Introduction to Financial Option Valuation: Mathematics, Stochastics and Computation, by Desmond J. Higham, Cambridge University Press, 2004.